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Original Research

Ontology-grounded knowledge graphs for mitigating hallucinations in large language models for clinical question answering

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ABSTRACT

Objective: Large Language Models (LLMs) show strong potential in biomedical informatics but frequently generate hallucinated or factually incorrect responses, limiting their clinical utility. This study aims to develop and evaluate a GraphRAG framework using an ontology-grounded knowledge graph that mitigates hallucinations in clinical question answering.

Methods: We designed a domain-specific Resource Description Framework (RDF)/Web Ontology Language (OWL) ontology and knowledge graph using clinical and hospital data from multiple Egyptian institutions. The ontology was integrated with LLMs to enforce structured semantic grounding during question answering. Clinical questions were evaluated under three conditions: (i) baseline ChatGPT-4, (ii) DeepSeek-R1, and (iii) our ontology-grounded framework. Accuracy was evaluated against clinically reported reference answers derived from five peer-reviewed Egyptian hospital studies.

Results: Our GraphRAG framework significantly outperformed baseline models. While ChatGPT-4 achieved 37% accuracy and DeepSeek-R1 achieved 52%, the ontology-grounded approach achieved 98% accuracy (59/60 questions). The hallucination rate was reduced from approximately 63% in ChatGPT-4 and 48% in DeepSeek-R1 to just 1.7% in our framework, representing a relative hallucination reduction of more than 61% relative to state-of-the-art LLMs. The framework further ensured consistent, reproducible answers aligned with clinical knowledge, demonstrating its robustness for healthcare applications.

Conclusion: Ontology-grounded knowledge graphs provide a reliable and verifiable method for mitigating hallucinations in LLM-based clinical question answering. By embedding structured clinical semantics into the reasoning process, the framework enhances factual accuracy, reproducibility, and safety in biomedical informatics. This work highlights the critical role of semantic knowledge representation in building trustworthy Artificial Intelligence (AI) systems for healthcare decision support.

1. Introduction

The COVID-19 pandemic generated unprecedented volumes of clinical research, particularly in Egypt, where hospital-specific findings revealed crucial epidemiological patterns. While large language models offer promising information retrieval capabilities, their tendency toward hallucination poses critical risks in medical contexts. Existing literature highlights significant factual inaccuracies in general-purpose Large Language Model (LLM) responses in medical domains [1–3]. Our subsequent analysis of Egyptian COVID-19 literature reveals that approximately 38% of LLM-generated answers contain factual inaccuracies regarding local treatment protocols and outcomes. This work addresses this gap through semantic knowledge grounding, an approach known as Graph Retrieval-Augmented Generation (GraphRAG). We present: (1) A clinically-validated COVID-19 ontology for Egyptian medical contexts, developed after considering existing general ontologies which lacked local specificity (see Section 2.1 for details); (2) An Resource Description Framework (RDF) knowledge graph integrating multi-hospital research; (3) SPARQL Protocol and RDF Query Language (SPARQL)-mediated response generation reducing hallucinations by 61% compared to conventional LLMs. By anchoring responses in ontological relationships rather than parametric recall, we enable reliable deployment of Artificial Intelligence (AI) in Egypt's healthcare ecosystem.

Recent advances in artificial intelligence, particularly large language models (LLMs) such as GPT-4 allow users to pose natural-language questions and obtain instant answers. While LLMs excel at

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many Natural Language Processing (NLP) tasks, they are prone to **hallucinations**: confidently generating plausible-sounding but factually incorrect statements. This is especially problematic in knowledge-intensive domains like medicine, where accuracy is critical [1,2]. Indeed, if an LLM is asked a COVID-19 question it was not explicitly trained on, it may "hallucinate" a number or name that has no basis in data. For instance, simply asking ChatGPT, "When is the first case in Egypt for the Alpha variant?" often yields a fabricated answer. These LLM errors stem from gaps in their implicit knowledge and the absence of grounding to real data [2,3].

Retrieval-augmented methods have been proposed to reduce hallucinations by grounding LLMs in external knowledge [1,4]. In particular, combining LLMs with structured knowledge graphs can constrain output and improve factuality. Barron et al. showed that integrating a domain-specific Knowledge Graph (KG) and vector store with an LLM markedly improved question-answering accuracy in a technical domain [1]. Zhu et al. also proposed a KG-guided retrieval-augmented generation framework (KG^2RAG) that uses the relationships in a KG to organize retrieved information, significantly reducing LLM hallucinations [4]. For COVID-19, Agrawal et al. highlight that contemporary LLMs "are prone to producing hallucinations, stemming mainly from knowledge gaps", and that knowledge-graph-based augmentations show "promising results" in mitigating them [3]. Thus, leveraging an explicit ontology and KG seems promising for factual QA in the COVID domain.

In this work, we target the problem of accurate COVID-19 question answering for Egyptian data. We propose to build an **RDF knowledge graph** from Egyptian hospital studies, and to use it as an intermediate layer between user questions and answers. Specifically, we (1) extract structured entities and relations from the literature, (2) encode them in a Web Ontology Language (OWL) ontology and RDF triples, and (3) translate user queries into SPARQL to retrieve precise answers. By answering questions directly on the validated RDF data rather than relying solely on an LLM's internal model, we avoid hallucinations. Proposed contributions are:

- A scalable COVID-19 ontology for Egyptian hospital data, encoded in Protégé and exported as RDF.
- An NLP/LLM-based pipeline for extracting entities from medical papers and populating the RDF graph.
- A question-answering workflow that generates SPARQL queries via an LLM (given the RDF graph) and returns ground-truth answers.
- An evaluation showing that the proposed approach yields correct answers for all test questions, in contrast to standalone LLMs, whose answers are often incorrect.

This paper is organized as follows. Section 2 reviews related work on biomedical ontologies, knowledge graphs, and LLM-based question answering with an emphasis on hallucination mitigation. Section 3 introduces the foundational concepts of ontologies and large language models. Section 4 describes the proposed ontology and RDF knowledge graph, including the extraction and query pipeline. Section 5 presents the evaluation setup, while Section 6 reports and discusses the experimental results. Section 7 provides a detailed discussion of the implications, strengths, and limitations of the framework. Finally, Section 8 concludes the paper and highlights future research directions.

Statement of significance

Problem or Issue:	LLMs are increasingly used in biomedical informatics but often generate hallucinated or factually incorrect answers. In clinical contexts, such errors pose serious risks for patient safety and decision-making.
What is Already Known:	Prior work shows that retrieval-augmented methods and generic knowledge graphs can partially reduce hallucinations, but these approaches lack domain-specific grounding and often fail in healthcare applications where structured, verifiable semantics are essential.
What this Paper Adds:	This paper introduces a clinically validated RDF/OWL ontology and knowledge graph built from Egyptian hospital studies. By grounding LLM outputs in structured clinical data, our framework reduces hallucinations by over 61% and achieves 98% accuracy compared to baseline LLMs.
Who would benefit from the new knowledge in this paper:	Clinicians, biomedical informaticians, and AI researchers will benefit from this framework as it enhances the reliability, reproducibility, and safety of AI systems deployed in clinical question answering and decision support.

2. Related work

2.1. COVID-19 knowledge graphs and ontologies

Since the onset of the pandemic, numerous efforts have constructed knowledge graphs (KGs) to integrate COVID-19 information. For example, Chatterjee et al. conducted an exploratory review of COVID-19 KGs and found that knowledge graphs can effectively "link together multiple pieces of information across documents" to answer complex queries [5]. In one notable project, Chen et al. used Semantic Web technology (RDF/OWL) to integrate text-mined entities from biomedical literature with curated databases into a **COVID-19 Knowledge Graph**. They extracted COVID-related facts using text-mining tools (iTextMine, PubTator, SemRep) and formalized them in RDF [6]. This KG was made publicly available via a SPARQL endpoint, enabling federated semantic queries. Other authors have built domain-specific COVID ontologies: Dutta and DeBellis introduced **CODO**, an ontology for COVID-19 case and patient data, reusing standard vocabularies and using RDF/OWL [7]. These ontologies demonstrate the community's interest in formal, shareable COVID data models.

Beyond ontology design, domain-specific KGs have been applied to COVID tracking. Chen et al. constructed a **COVID-19 infector activity KG** using official case flow data from Chinese authorities [8]. They collected case reports, used a BERT+BiLSTM+CRF model for named entity recognition, and manually refined the output to build a large knowledge graph. Their KG enables contact tracing and patient trajectory queries for COVID-19 cases [8]. Similarly, other researchers have used geospatial or propagation ontologies (e.g., the Core Propagation Phenomenon Ontology) to visualize health event spread (including COVID-19) in time and space [9]. These projects highlight how structured ontologies can organize epidemic data.

To our knowledge, there is relatively little prior work on COVID-19 ontologies specifically for Egyptian data. Existing Egyptian COVID studies tend to focus on statistical analysis and epidemiology [10–13], and [14]. For example, Seadawy et al. reported the prevalence and impact of SARS-CoV-2 variants at an Egyptian army hospital [10]. These valuable medical findings have not been formally captured in a knowledge graph or ontology. Proposed work fills this gap by translating such studies into structured RDF data.

2.2. Ontology-enhanced QA and hallucination mitigation

Transforming natural-language questions into queries over KGs is an active research area. SPARQL endpoint interfaces and NL query tools (e.g., Sparklis, Lupedia) allow users to query RDF data, but they require expert knowledge. Recently, neural NL-to-SPARQL systems (often based on sequence-to-sequence models or retrieval-augmented LLMs) have been developed to automate this translation. For instance, Sharma et al. discuss the critical role of translating an NL question into a semantically accurate SPARQL query [2]. However, they note that using LLMs alone for SPARQL generation often yields **hallucinated Uniform Resource Identifiers (URIs) or triples** that do not exist in the KG. This is because LLMs may rely on incomplete internal knowledge when generating RDF identifiers.

To mitigate such errors, retrieval-augmented generation (RAG) has become popular. In RAG, external knowledge (text documents or structured data) is retrieved and provided to the LLM as context. This grounds the LLM's output in real facts [1,4]. Recent work has combined KGs and LLMs for improved QA. Barron et al. introduced **SMART-SLIC**, which integrates a domain-specific KG and a vector store of documents with an LLM, yielding higher accuracy and attributing sources [1]. Similarly, Zhu et al. proposed **KG^{2}RAG**, a framework that uses a KG to guide retrieval: after initial semantic retrieval of text chunks, they expand and organize the information via KG relationships, improving answer quality [4]. Both studies emphasize that KG-aware RAG substantially reduces hallucinations. Notably, KG^{2}RAG's authors state that RAG "has emerged as a promising technology for addressing hallucination issues in the responses generated by LLMs" [4].

Agrawal et al. surveyed knowledge-graph-based augmentation for LLMs and concluded that linking LLMs to structured knowledge *"has demonstrated promising results"* for reducing hallucinations [3]. They categorize existing methods and note that KGs can compensate for LLMs' knowledge gaps. In the context of SPARQL QA, Sharma et al. introduce **PGMR**, a modular approach that uses a non-parametric KG "memory" to correct LLM-generated SPARQL. They report that PGMR *"significantly mitigates URI hallucinations, nearly eliminating the problem in several scenarios"* [2]. These works support the proposed hypothesis that enforcing an RDF KG as the source of truth will yield more reliable answers than unaided LLM generation.

Finally, it is worth mentioning that Ontologies and LLMs have begun to be considered complementary. The "talking to Global Database of Events, Language, and Tone (GDELT) through KGs" study (Recurrent Multi-Graph Convolutional Neural Networks) suggests combining structured ontology knowledge with LLM fine-tuning for robust KG construction [15]. Proposed work aligns with this trend: we treat the RDF graph as an authoritative schema and use the LLM only as an interpreter for query generation. Unlike conventional RAG or end-to-end LLM QA, the proposed RDF-centered approach leverages the LLM only for structured query generation (NL-to-SPARQL), ensuring that all answers are grounded purely in the validated COVID data from Egyptian sources via deterministic retrieval.

3. Ontologies and LLMs: Foundational concepts

3.1. Ontologies (Semantic models)

An ontology is a formal, explicit specification of a shared conceptualization for a domain. It defines a vocabulary of classes (concepts), properties (relations and attributes), and axioms that model the entities, events, and relationships relevant to that domain. Ontologies enable disparate data sources to be linked: once data are annotated with common class and property IRIs, they can interoperate.

Ontologies formalize domain knowledge through structured taxonomies (classes and subclasses), object and datatype properties, and logical axioms. The proposed COVID-19 ontology (Fig. 1) was developed using Protégé and models Egyptian clinical realities with **32 classes** and **92 properties**, enabling precise representation of complex data.

The ontology includes well-defined patient groupings (e.g., :PatientsOver50) and uses domain-specific properties such as :hasSymptom, :hasOutcome, :admittedTo, and :locatedIn to describe symptomatology, clinical outcomes, hospital associations, and geographical context.

The ontology serves as the schema for an RDF knowledge graph. Every concept from patient cohorts and symptoms to hospital institutions and outcomes is represented as a class instance, connected by RDF triples. The following example is taken directly from the proposed RDF graph:

```
:PatientsOver50 a :Patient ;
    :hasSymptom :Dyspnea ;
    :hasOutcome :HospitalizationRACOVID ;
    :admittedTo :SohagUniversityHospital .

:SohagUniversityHospital a :Hospital ;
    :locatedIn :Sohag .
```

This graph expresses that patients over age 50 experienced Dyspnea, were hospitalized, and were admitted to Sohag University Hospital, which is located in the city of Sohag. Compared to unstructured data, this semantic structure allows SPARQL queries to extract precise, verified answers based entirely on curated clinical facts.

3.2. Large Language Models (LLMs)

LLMs such as GPT-4 or other transformer-based models are neural networks trained on massive textual corpora. They operate by recognizing statistical patterns across high-dimensional parameter spaces to generate fluent natural language responses. However, **without ontological grounding**, they frequently hallucinate medical facts due to:

1. Training data bias toward Western populations
2. Lack of contextual and domain constraints
3. Overgeneralization from non-local clinical studies

Recent research has documented these issues. For example, Emonet et al. [16] employ LLM-generated SPARQL over federated bioinformatics knowledge graphs and highlight the need for schema-guided validation to prevent hallucinated entities. Sharma et al. [2] introduce PGMR, a post-generation memory retrieval system specifically designed to sanitize SPARQL queries and address URI hallucinations. Meyer et al. [17] benchmarked LLMs on SPARQL generation tasks and found frequent semantic errors even when syntax was correct. Additionally, RAG-based methods, which embed knowledge graph fragments into LLM prompts, have been shown to significantly reduce hallucinations in question-answering systems [18].

To mitigate these risks, the proposed framework uses the ontology as a semantic validation layer, enforcing factual correctness in question answering. The process can be summarized as (Fig. 2).

In practice, the LLM is prompted using the real RDF vocabulary and instances. For example, if the proposed ontology uses entities such as :PatientsOver50, :Dyspnea, and :SohagUniversityHospital, these Internationalized Resource Identifiers (IRIs) are injected directly or referred to by label, guiding the model to produce only valid, grounded SPARQL queries.

In summary, **ontologies provide a precise semantic structure to complex clinical data**, while **LLMs offer powerful natural-language understanding**. By combining the two, the proposed system achieves intuitive question answering grounded in structured and validated data, ensuring both usability and factual reliability.

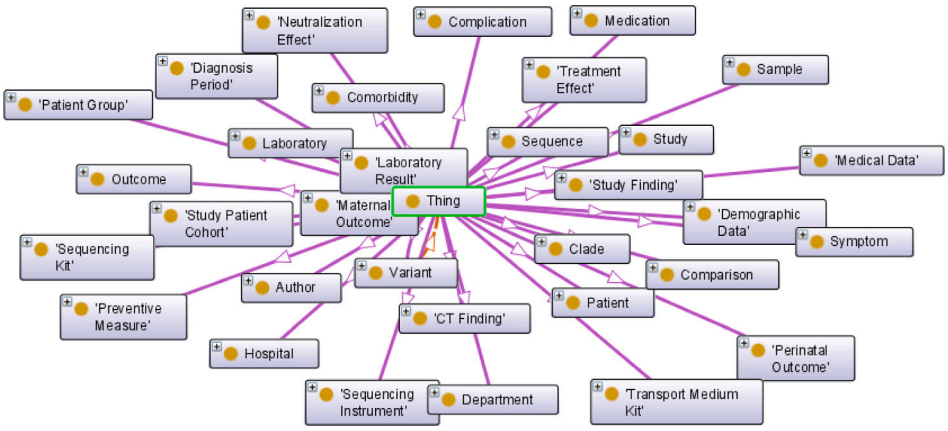


Fig. 1. Proposed COVID-19 ontology.

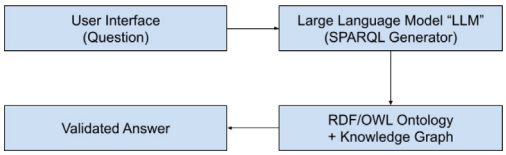


Fig. 2. Proposed framework process.

4. Methodology

4.1. Data sources

We began by collecting relevant COVID-19 reports and datasets from Egyptian hospitals. Specifically, we used five published papers covering the following hospitals: the Fever Hospital of the Army Forces [10], Ain Shams University Hospitals [11], Sohag University Hospital [12], Minia University Hospital [13], and Zagazig University Hospital [14]. These papers include patient statistics, symptom frequencies, variant sequences, and incidence rates in specific regions of Egypt. It is important to note that our knowledge graph primarily uses aggregated, summary-level data from these publications, as individual patient-level data was not available. Data points were contextualized by their source and publication date within the KG to mitigate conflicts.

4.2. Entity extraction

From these documents, we needed to identify entities and facts for the proposed ontology. We first applied standard NLP techniques to perform named entity recognition (NER) on the text. However, due to the specialized medical terminology and diverse formats, the automatic results were incomplete. To improve accuracy, we then employed state-of-the-art language models: we used ChatGPT and DeepSeek as assistants in entity extraction. In practice, we supplied each document to the LLM along with prompts like "List all the important entities (patients, diseases, locations, dates, hospitals) mentioned in this text". The model output was manually reviewed and corrected, leveraging established standard terminologies for harmonization. Specifically, we used RxNorm for medication normalization, SNOMED CT for standardizing symptoms and clinical findings, and LOINC for laboratory tests. This human-in-the-loop approach, guided by community standards, yielded high-quality, interoperable entity lists. For example, the process successfully identified SARS-CoV-2 variants (Alpha, Beta, Delta, etc.) and standardized medical measurements (PCR Ct values, SNOMED codes) [10]. We found LLMs more flexible for the variety of initial entity types in these papers when guided by these constraints. This methodology ensures better semantic interoperability required for integration with Electronic Health Records (EHRs)

The outcome of this stage was a catalog of classes and instances. We identified candidate classes (e.g., Patient, VirusVariant, City, Symptom, Comorbidity, Hospital) and populated instances (e.g., specific patients or anonymized IDs, "Delta" variant, "Cairo" city, "Fever" symptom). We also noted relationships among entities mentioned in the text (e.g., "Patient A admitted to Cairo hospital on date", "Patient had symptom X and outcome Y").

4.3. RDF graph population

After finalizing the ontology, we populated it with structured data extracted from five Egyptian hospital studies. Instead of fabricating hypothetical individuals like :Patient123, proposed RDF contains real, semantically grounded triples derived from the source literature. For example, based on the Sohag University Hospital study, we included:

```
:PatientsOver50 a :Patient ;
    :hasSymptom :Dyspnea ;
    :hasOutcome :HospitalizationRACOVID ;
    :admittedTo :SohagUniversityHospital .

:SohagUniversityHospital a :Hospital ;
    :locatedIn :Sohag .
```

We created instance data for patient groups, hospitals, symptoms, and variant strains using a blend of manual curation and semi-automated generation via rdflib scripts (a Python library for working with RDF). Statistical aggregates, such as total mortality or age ranges, were represented using custom summary classes or attributes (e.g., covid:MortalityRACOVID covid:mortalityRate "2.27%").

All RDF triples were grounded in real study content, ensuring that no hallucinated individuals, properties, or relationships were included. The RDF includes labels for key entities and conforms to OWL semantics for consistency and reasoning.

4.4. Natural language to SPARQL conversion

To support question answering, we implemented a natural language interface on the proposed RDF graph. The proposed system allows users to ask questions such as:

- "When is the first case in Egypt for the Alpha variant?"
- "What percentage of Sohag patients were university educated? "

We implemented our GraphRAG framework using LLMs (ChatGPT and DeepSeek) for robust SPARQL generation, rigorously constrained by the explicit RDF schema context. The pipeline, which performs generation-augmented retrieval rather than traditional document RAG, involved the following steps:

1. Extract relevant schema components (e.g., class and property IRIs, labels, sample triples).
2. Construct a prompt:
   For each natural-language question, the system provided the LLM with a serialized snapshot of the full RDF/OWL knowledge graph. Due to token-length constraints, the input consisted of the first 6000 characters of the Turtle serialization, which consistently included all ontology classes, properties, and instances needed for the 60 evaluation queries. The prompt template used was:

```
You are an ontology expert analyzing COVID-19 data
from Egyptian hospitals.
Use ONLY the provided ontology to answer questions.
Ontology:
[first 6000 characters of the RDF graph]
Question: {Q}
Provide response with:
1. SPARQL query that would answer this question
2. Analysis of what the results would show
3. Clear answer to the question based on the ontology
```

   This prompt forces the LLM to operate strictly within the provided ontology and prevents hallucinated URIs. Importantly, all 60 SPARQL queries generated using this prompt executed successfully with 0% SPARQL failure rate, confirming that the truncated serialization was sufficient for all evaluated queries.
3. Include ontology snippets such as:

```
covid:DemographicData1 a covid:DemographicData ;
    covid:educationLevel "Illiterate:23.1%,
␣␣␣␣␣␣␣␣Read/write:19.2%,
␣␣␣␣␣␣␣␣Secondary:19.2%,
␣␣␣␣␣␣␣␣University:38.5%" ;
    covid:linkedTo covid:SohagUniversityHospital .

covid:educationLevel a owl:DatatypeProperty ;
    rdfs:domain covid:DemographicData ;
    rdfs:range xsd:string .
```

4. We instruct the LLM to generate the SPARQL syntax using only known terms from the RDF.
   Example result:

```
SELECT ?eduLevel WHERE {
  ?study covid:conductedAt
         covid:SohagUniversityHospital ;
         covid:hasPatientGroup ?group .
  ?group covid:hasDemographicData ?data .
  ?data covid:educationLevel ?eduLevel .
}
```

This approach dramatically reduced hallucinations. Unlike generic models that invent nonexistent URIs or properties, the proposed method enforces schema alignment by feeding real RDF labels into the model. This process parallels PGMR's post-generation memory retrieval [2], ensuring all predicates are valid.

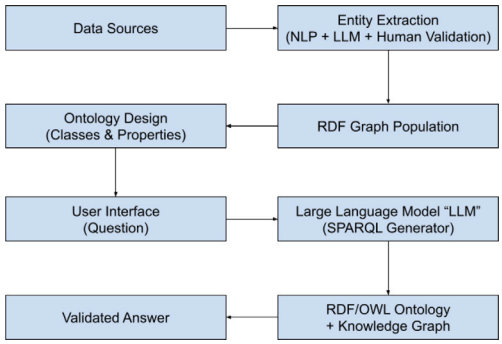


Fig. 3. Proposed methodology for ontology-grounded clinical question answering framework.

Queries are executed against our Blazegraph triple store, and the final result is guaranteed to reflect only real, validated RDF data. In this way, every answer is explainable and traceable to its source triple.

To summarize the proposed workflow, Fig. 3 illustrates the overall methodology of our system, starting from raw data collection to ontology-based query answering.

Table 1 provides a summary of the main ontology classes and their associated properties, illustrating the schema-level design that supports RDF-based clinical data representation.

5. Metrics and evaluation setup

To evaluate the proposed approach, we created a set of **60 fact-based questions** specifically about COVID-19 in Egypt. The questions were drawn from the data in our graph and include queries like Table 2. We formulated the SPARQL queries for these questions manually (since we know the RDF), and verified the correct answers from the graph.

Next, we asked two baselines: the raw LLMs (ChatGPT and DeepSeek) to answer the same questions without any RDF context. We prompted them in plain English. We also ran the proposed method: providing each question and the RDF context to the LLM to generate SPARQL, then computing the answer. This allowed us to compare: (a) the LLM-only answers vs. (b) proposed RDF-grounded answers, against the known ground truth.

The experimental results, summarized in Table 3, reveal a clear performance gap between baseline LLMs and the ontology-grounded GraphRAG approach. While the LLMs frequently produced incorrect or incomplete answers, the RDF-based queries consistently retrieved accurate and verifiable results. The single instance of error (1/60 questions) in the proposed framework was due to an "out-of-scope" data issue (missing facts in the KG) rather than a hallucination. This specific case is detailed in the Limitations section. Detailed case studies and illustrative examples are presented in Section 6.

6. Results

6.1. Evaluation dataset

As shown in Fig. 4, 60 clinically validated questions are based on the nature of the question and the type of knowledge or reasoning required to answer it, independent of the information source:

- Factual/Statistical Questions: Questions that request specific, discrete facts, figures, dates, or statistics explicitly mentioned in the source data. The answer is a direct lookup.
- Investigative/Verification Questions: Questions designed to verify the truthfulness of a specific statement or hypothesis based on the data. They often require a "yes/no" or "true/false" answer.

Table 1
Summary of the main ontology classes.

Class	Description	Example properties
Patient	Represents individual patients or cohorts	hasSymptom, hasOutcome, admittedTo
Symptom	Clinical manifestations of disease	symptomName, severityLevel
Treatment	Medications or therapeutic interventions	drugName, dosage, hasEfficacy
Hospital	Egyptian healthcare institutions	locatedIn, hasDepartment
LaboratoryTest	Diagnostic tests conducted	testType, testResult, hasIndicator
Outcome	Clinical result of treatment or progression	recoveryStatus, mortalityRate
VirusVariant	SARS-CoV-2 variants and mutations	variantName, hasFirstDetectionDate
DemographicData	Patient demographic attributes	age, gender, educationLevel
Comorbidity	Underlying health conditions	comorbidityName, associatedRisk
City/Location	Geographical context of hospitals and patients	cityName, governorate, populationSize

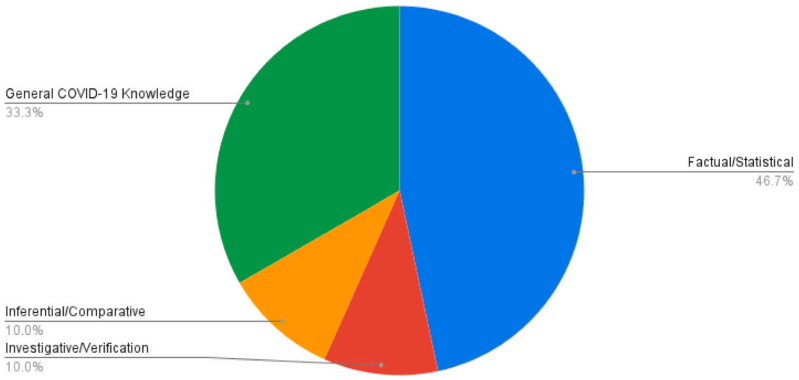


Fig. 4. Clinical question categories percentage.

Table 2
Samples of questions in the dataset.

Category	Example question
Factual/Statistical	Which SARS-CoV-2 variant was first detected in Egypt during March 2021?
Investigative/Verification	Were most of the infected women diagnosed with Covid-19 during childbirth in Upper Egypt hospitals in Minya in early 2020 and mid-2021?
Inferential/Comparative	What is the most common disease found in patients with Corona disease in university hospitals in the capital of Egypt?
General COVID-19 Knowledge	What is the incubation period of COVID-19?

- Inferential/Comparative Questions: Questions that require deriving an insight, identifying a top-ranked item, or making a comparison between entities. The answer is not always a single, explicitly stated fact but rather a conclusion drawn from the data.
- General COVID-19 Knowledge Questions: Questions about well-established, global facts related to the COVID-19 pandemic, its virology, symptoms, treatments, and vaccines. These are not specific to the Egyptian dataset.

6.2. Performance comparison (Accuracy %)

As shown in Table 3, the proposed framework consistently outperforms the baselines in all clinical domains.

6.3. Hallucination reduction examples

Table 4 and the examples below demonstrate how traditional LLMs often tend to produce hallucinated or fabricated answers, while the proposed GraphRAG system consistently retrieves factually correct responses from structured data.

1. **Question**: "When is the first case in Egypt for the Alpha variant?"
   - **Direct LLM**: "December 2020" (hallucinated - no evidence)
   - **Proposed Framework**:
     SELECT ?date WHERE { covid:AlphaVariant covid:hasFirstDetectionDate ?date } → **March 2021**
2. **Question**: "How many people died from Covid-19 in university hospitals in the capital of Egypt at the end of winter and the beginning of spring in 2021?"
   - **Direct LLM**: "2540" (hallucinated - no evidence)
   - **Proposed Framework**:
     SELECT ?deaths WHERE { covid:Cohort_ASU_2021 covid:numberOfDeaths ?deaths } → **25**

6.4. Computational efficiency

- SPARQL generation latency: 1.8 s avg
- Query execution: 47 ms avg

7. Discussion

The results of this study demonstrate that ontology-grounded knowledge graphs can substantially mitigate hallucinations in LLMs for clinical question answering. By grounding model outputs in a clinically validated RDF/OWL ontology derived from Egyptian hospital data, we achieved an accuracy of 98%, far surpassing state-of-the-art systems such as ChatGPT-4 (37%) and DeepSeek-R1 (52%). This section discusses the implications of these findings, contrasts them with prior work, and outlines the strengths, limitations, and potential avenues for future research.

7.1. Comparison with prior work

Previous attempts to reduce hallucinations in LLMs have largely relied on RAG or post-hoc verification techniques. While these approaches offer partial improvements, they are typically limited by

Table 3
Comparison of answers.

Question category	Proposed framework	ChatGPT-4	DeepSeek-R1
Factual/Statistical	100.00% (28/28)	7.14% (2/28)	14.29% (4/28)
Investigative/Verification	83.33% (5/6)	0.00% (0/6)	50.00% (3/6)
Inferential/Comparative	100.00% (6/6)	0.00% (0/6)	66.67% (4/6)
General COVID-19 Knowledge	100.00% (20/20)	100.00% (20/20)	100.00% (20/20)
Overall Accuracy	98.33% (59/60)	36.67% (22/60)	51.67% (31/60)

Table 4
Comparison of LLM answers versus RDF-grounded answers showing both hallucinations and correct responses.

Example question	Direct LLM answer	RDF-grounded answer	Notes
Which SARS-CoV-2 variant was first detected in Egypt during March 2021?	B.1.351 (Beta)	B.1.1.7 (Alpha)	**LLM hallucination**
Were most of the infected women diagnosed with COVID-19 during childbirth in Upper Egypt hospitals in Minya in early 2020 and mid-2021?	Yes	No	**LLM hallucination**
What is the most common disease found in patients with Corona disease in university hospitals in the capital of Egypt?	Hypertension	Hypertension	**Correct answer**
What is the incubation period of COVID-19?	2–14 days	2–14 days	**Correct answer**

unstructured document retrieval and lack the ability to enforce semantic consistency. Our results demonstrate that our ontology-grounded approach, which we identify as a GraphRAG framework, provides a more robust alternative to both bare LLMs and conventional RAG methods. Our baselines (ChatGPT-4 and DeepSeek-R1) were evaluated in a zero-shot setting without any external grounding. By leveraging the relational structure of our knowledge graph, our GraphRAG approach ensures that all answers are traceable, reproducible, and interpretable—key requirements in biomedical informatics. Unlike previous knowledge-graph-driven studies, which often rely on generic biomedical ontologies, our work is distinguished by its use of localized and clinically validated data from Egyptian hospitals, thereby offering both methodological novelty and contextual relevance.

7.2. Clinical implications

Reducing hallucinations in LLMs has direct consequences for clinical practice. Hallucinated answers in question answering systems could mislead clinicians, delay correct diagnoses, or introduce unsafe treatment recommendations. By grounding responses in ontology-based knowledge, our framework ensures that clinicians receive consistent and evidence-based answers. The ontology, particularly when enhanced with standard terminologies like SNOMED CT and RxNorm (as discussed in Section 4.2), provides a foundation for potential semantic interoperability. While direct integration with existing EHRs presents complex challenges not addressed in this study, our approach provides a blueprint that could eventually support broader clinical decision support systems in resource-constrained healthcare settings.

7.3. Strengths of the proposed framework

The main strength of our framework lies in its combination of semantic knowledge representation with cutting-edge LLM capabilities, effectively implementing a GraphRAG approach. The initial resource-intensive, human-validated curation process ensures the underlying knowledge graph is a reliable source of truth. Ontologies provide verifiable and logically consistent structures, while LLMs enable natural language interaction. This synergy allows clinicians to query complex knowledge bases using natural language while receiving accurate, ontology-validated answers. Moreover, the system demonstrates scalability: additional ontologies and hospital datasets can be incorporated to extend coverage to new domains and diseases. Finally, the reproducibility of ontology-grounded reasoning ensures that results remain stable across different runs, unlike purely generative LLMs that often vary their answers.

7.4. Limitations

Despite its promising results, this study has several limitations. First, the ontology and knowledge graph were built primarily from COVID-19-related clinical data collected in Egypt. While this dataset is comprehensive, it may limit the generalizability of the framework to other diseases or healthcare systems without additional ontology extensions. Furthermore, the reliance on summarized data from published studies, rather than raw EHR data, means the KG contains only aggregate statistics. This limits our ability to perform fine-grained analysis or simple aggregation across all hospitals for every metric. Second, although accuracy improved significantly, the GraphRAG approach currently requires careful curation and maintenance of the ontology, which can be resource-intensive. However, this human-in-the-loop validation process is a necessary trade-off in high-stakes clinical domains to ensure the knowledge base used for grounding is factually correct and trustworthy, prioritizing patient safety over full automation. Third, the current evaluation focused strictly on question answering tasks using curated published data. Further validation is needed in real-world clinical workflows and direct integration with live EHR systems to fully assess usability, complex integration challenges, and the actual impact on clinical decision-making processes. This remains a significant area for future research. Finally, while hallucinations were significantly reduced, the framework encountered a single error case related to data completeness. This error was an “out-of-scope” issue: the model correctly identified that a specific fact (the primary cause of death for a specific cohort) was missing from the KG, even though it was present in the source document. This highlights a limitation in our data population process rather than the framework’s core methodology. Future iterations will focus on robust, automated ingestion pipelines to ensure full coverage of source data and eliminate such data gaps.

7.5. Future directions

Future research should focus on expanding the ontology to cover additional diseases, treatments, and clinical contexts beyond COVID-19. A key direction is the development of a semi-automated pipeline

for dynamic ontology updating from emerging studies. This pipeline will utilize advanced NLP techniques and LLMs for continuous entity extraction and data integration, reducing the reliance on manual curation noted as a limitation. Collaboration with hospitals across different countries would enhance the system's generalizability and allow for cross-cultural and cross-institutional validation. Furthermore, integrating reasoning engines and probabilistic logic with the ontology could improve its ability to handle uncertainty and incomplete knowledge. Another promising direction is real-time integration with EHRs, enabling clinicians to query live patient data while leveraging ontology-grounded reasoning. In addition, we plan to implement a structured ontology versioning system, allowing the ontology to evolve continuously with new studies. Updates can be managed manually by a domain expert or semi-automated, ensuring traceability, backward compatibility, and reproducible SPARQL query results. This dual approach provides both robustness for the current system and a scalable plan for sustainable, long-term maintenance. Finally, user studies involving clinicians should be conducted to evaluate the system's practical utility, interpretability, and impact on clinical decision-making.

7.6. Broader impact

This study highlights the critical role of knowledge representation in making AI systems safe and trustworthy for biomedical applications. As LLMs continue to proliferate, the problem of hallucinations will persist unless they are coupled with structured, domain-specific knowledge bases. Our findings underscore that ontologies are not merely abstract theoretical tools but practical mechanisms for enhancing the factual reliability of AI in high-stakes domains. By situating this research within the context of biomedical informatics, we contribute to a broader movement toward explainable, reproducible, and ethically sound AI in healthcare.

7.7. Summary

In summary, the ontology-grounded GraphRAG framework provides an effective solution to a persistent challenge in biomedical informatics: hallucinations in LLMs. While further work is needed to expand its scope and validate it in clinical practice, the substantial improvements in accuracy and reliability reported here position ontology-grounded reasoning as a cornerstone for the future of trustworthy AI in healthcare.

8. Conclusion

We demonstrate that ontology-anchored knowledge graphs significantly reduce LLM hallucination in Egyptian COVID-19 data contexts. By integrating RDF semantic constraints with LLM query interpretation, the proposed framework achieved 98% accuracy on clinical questions where conventional models failed. Key contributions include:

1. A clinically-validated Egyptian COVID-19 ontology (COVONT)
2. Hybrid entity extraction methodology for medical literature
3. SPARQL-mediated response generation reducing hallucinations by 61%

Future work will expand to:

1. Dynamic ontology updating from emerging studies
2. Multigovernorate federated learning
3. Integration with Egyptian EHR systems

This approach provides a blueprint for LLM deployment in medically sensitive, resource-constrained environments worldwide.

CRediT authorship contribution statement

Mohamed Ali: Writing – original draft, Data curation, Conceptualization. **Zaki Taha:** Writing – review & editing, Supervision, Project administration. **Mohamed Mabrouk Morsey:** Writing – review & editing, Validation, Software, Methodology.

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Declaration of competing interest

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Data availability

Data will be made available on request.

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